WORKED SOLUTION

STEP III 2016, Question 12

Probability and Statistics • Chebyshev's Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q12

The key idea

Chebyshev's inequality bounds the probability of being *far* from the mean, so a statement that something is *close* to the mean comes from the complement. In each part the job is to spot the right random variable, find its mean and standard deviation, and choose k so that $k\sigma$ is exactly the distance we care about. In part (ii) the sum $\sum n^r/r!$, sitting next to e^n, is the fingerprint of a Poisson distribution.

Solution

Part (i)

Let X be the number of heads in the $100n$ throws. Assuming the throws are independent, each landing heads with probability 0.2, we have $X \sim \mathrm{B}(100n, 0.2)$, so

$$\mu = 100n \times 0.2 = 20n, \qquad \sigma^2 = 100n \times 0.2 \times 0.8 = 16n, \qquad \sigma = 4\sqrt{n}.$$

The event $16n \le X \le 24n$ is exactly $|X - 20n| \le 4n$, so

$$\alpha = \mathrm{P}(|X - \mu| \le 4n) = 1 - \mathrm{P}(|X - \mu| > 4n).$$

We want $k\sigma = 4n$, that is $k \cdot 4\sqrt{n} = 4n$, so $k = \sqrt{n}$, which is positive. Chebyshev's inequality gives

$$\mathrm{P}(|X - \mu| > 4n) = \mathrm{P}(|X - \mu| > \sqrt{n}\,\sigma) \le \frac{1}{(\sqrt{n})^2} = \frac{1}{n},$$

and therefore

$$\alpha \ge 1 - \frac{1}{n},$$

as required. (The condition $n > 1$ just makes the bound say something: for $n = 1$ it reads $\alpha \ge 0$.)

Part (ii)

Ask yourself Which random variable has probabilities that look like $n^r/r!$?

If $Y \sim \mathrm{Po}(n)$ then $\mathrm{P}(Y = r) = \mathrm{e}^{-n} n^r / r!$ for $r = 0, 1, 2, \ldots$, so

$$\mathrm{e}^{-n}\left(1 + n + \frac{n^2}{2!} + \cdots + \frac{n^{2n}}{(2n)!}\right) = \mathrm{P}(Y \le 2n).$$

The required inequality, after dividing by $\mathrm{e}^n > 0$, is therefore equivalent to

$$\mathrm{P}(Y \le 2n) \ge 1 - \frac{1}{n}.$$

For a Poisson distribution the mean and variance are both n, so $\mu = n$ and $\sigma = \sqrt{n}$.

Ask yourself Chebyshev's inequality is two-sided, but $Y \le 2n$ is a one-sided event. How do they match up?

Because $Y \ge 0$, we always have $Y - n \ge -n$, so $Y - n < -n$ is impossible. Hence

$$|Y - n| > n \iff Y - n > n \iff Y > 2n.$$

Taking $k = \sqrt{n}$, so that $k\sigma = n$, Chebyshev's inequality gives

$$\mathrm{P}(Y > 2n) = \mathrm{P}(|Y - n| > \sqrt{n}\,\sigma) \le \frac{1}{n},$$

so $\mathrm{P}(Y \le 2n) = 1 - \mathrm{P}(Y > 2n) \ge 1 - \frac{1}{n}$. Multiplying by e^n,

$$1 + n + \frac{n^2}{2!} + \cdots + \frac{n^{2n}}{(2n)!} \ge \left(1 - \frac{1}{n}\right)\mathrm{e}^n,$$

which is the result.

Checks

The bounds are true but generous. For $n = 2$ the exact binomial probability in (i) is $\alpha \approx 0.868$, against the bound 0.5; for $n = 10$ it is $\alpha \approx 0.9986$, against 0.9. In (ii) with $n = 2$ the left-hand side is $1 + 2 + 2 + \frac{4}{3} + \frac{2}{3} = 7$, while $\frac{1}{2}\mathrm{e}^2 \approx 3.69$. We checked both inequalities exactly for every n from 2 to 30 in (i), and from 1 to 59 in (ii).

Common slips

- Using the variance $16n$ in place of the standard deviation $4\sqrt{n}$ when choosing k.
- Forgetting that Chebyshev bounds the probability of being *outside* the interval, so we need the complement to get α.
- In (ii), not explaining why the lower tail $Y < 0$ contributes nothing.

Beyond STEP

Chebyshev's inequality is the simplest example of a *concentration inequality*, and applying it to an average of m independent trials gives the weak law of large numbers: the spread shrinks like $1/\sqrt{m}$. Part (ii) also says that the tail of the exponential series beyond $n^{2n}/(2n)!$ is at most e^n/n, which is an analytic fact proved by probability. Sharper tools such as Chernoff bounds show the true probabilities are exponentially close to 1. For more, see Grimmett and Welsh, *Probability: An Introduction*.

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